

# Frequency domain analysis of glottal flow: The LF-model revisited

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## 1. Introduction

A major tool for the study of the human voice source is inverse filtering. Over the years, a substantial amount of work in this area has been carried out at the Royal Institute of Technology (KTH) (Fant, 1993). Inverse filtering is a process of removing the vocal-tract transfer function from the speech wave thus regenerating a replica of the underlying source. This process provides us with some insight in the production mechanism and also a substance to be quantified and described within a suitable parametric system.

Early parametric systems concentrated on main shape aspects of glottal flow pulses such as rise time, decay time and open quotient. The importance of the flow discontinuity at closure as an excitation function was discovered at an early stage in connection with inverse filtering and was included in a Laplace transform production modelling in 1979 (Fant, 1979). Five years later the importance of the return phase in the flow derivative was fully acknowledged (Ananthapadmanabha, 1984) and became a major constituent of the LF-model (Fant—Liljencrants—Lin, 1985). The effective duration of the return phase,  $T_a$ , was shown to be inversely proportional to a frequency  $F_a = 1/2\pi T_a$  where the source spectrum attains an extra -6dB/oct slope. Increasing  $T_a$  thus implies a lowering of the cut-off frequency  $F_a$  of an equivalent low-pass filter. This parameter is usually of greater significance than the main pulse-shape parameters.

The ability to capture wave-shape essentials has promoted a wide use of the LF-model. However, human data from inverse filtering may deviate substantially from model data, mainly in terms of a superimposed fine structure which displays both recurrent patterns and a seeming randomness, see Fant (1982, 1986, 1993); Lin (1990); Fant—Lin (1988).

Systematic covariations in the LF parameters have been exploited in a transformed version (Fant—Liljencrants—Lin, 1985) of the model, which operates with fewer parameters retaining wave-shape essentials, and may be combined with a specification of the deviations of measured LF-parameters from predicted default values. This new system also has advantages from an

experimental point of view and is well suited for rule-oriented speech analysis and synthesis.

This data reduction-expansion is analogous to a principal component analysis, aiming at basic dimensions of the phonatory process. The choice of the peak value  $U_0$  of the oscillatory component of glottal flow, and the flow derivative  $E_e$  at the point of excitation together with  $F_0$  as main descriptive parameters, simplifies the inverse filtering process and the derivation of main glottal waveshape parameters. The significance of  $U_0$  is its close relation to the amplitude of the voice fundamental, while  $E_e$  is the basic determinant of formant amplitudes. Moreover, we have found systematic dependencies of  $E_e$  and  $U_0$  on  $F_0$  which are scaled differently for male and female voices (Fant—Kruckenberg, 1994).

A special study has now been devoted to frequency-domain methods of LF-parameter determination. There are two basically different systems that may be employed. One is the conventional inverse filtering system which, in addition to the time-domain matching, may employ a frequency domain check. The other method is analysis-by-synthesis, in which a spectral section of the recorded sound is compared with the output of a reference synthesizer, the parameter settings of which are varied to provide an optimal spectral match. Systematic differences in results with the two methods are related to different conventions for dealing with vocal-tract length and locating poles higher than  $F_4$ .

Major ambiguities exist in parameter determinations. There are subjective elements in time-domain matching, some of which have a correspondence in alternative choice of frequency regions to be matched. An interesting question is to what extent these ambiguities provide covariations that cancel out in the transformed version of the LF-model.

The covariation of source and filter functions - or in more general terms that of phonatory and articulatory processes - is of particular interest. It is the combined gesture rather than the source function alone which has a communicative function. Supraglottal constrictions impede the voice source (Bickley—Stevens, 1986) and glottal abduction introduces additional bandwidth and  $F_1$  increase. Subglottal coupling and aspiration noise add to the source features (Lin, 1990; Fant—Lin, 1988; Bickley—Stevens, 1986; Klatt—Klatt, 1990; Karlsson, 1992; Karlsson—Neovius, 1993; Gobl—Ní Chasaide, 1988; Stevens—Hanson, 1994).

A specific topic of interest in prosody is the coordination of glottal adjustments, adduction/abduction gestures,  $F_0$ -control, and lung pressure. There are apparent differences between singing and speech that need to be studied in greater detail. Applications in prosody are dealt with in other publications (Fant—Kruckenberg, 1994, 1995).

## 2. The transformed LF-model

The LF-model (Fant—Liljencrants—Lin, 1985) is illustrated in Figure 1. We have already discussed the significance of the return phase which accounts for the degree of spectral tilt through the  $F_a=1/(2\pi T_a)$  parameter which is frequently used as an alternative to  $R_a=T_a/T_0$ .

The  $R_k=(T_e-T_p)/T_p$  parameter specifies the relative duration of the falling branch from the peak at time  $T_p$  to the discontinuity point  $T_e$ .

The  $R_g=T_0/2T_p$  parameter increases with a shortening of the rise time  $T_p$ . A large  $R_g$  and a small  $R_k$  thus produce a small open quotient,  $OQ=(1+R_k)/2R_g$ . A common alternative definition is  $OQ'=(1+R_k)/(2R_g)+R_a$  which includes the return phase.

Typical values for male vowels (Gobl, 1988) are  $F_a=700$  Hz,  $R_k=0.30$ ,  $R_g=1.20$  and for female vowels (Karlsson, 1988, 1990; Fant, 1995),  $F_a=400$  Hz,  $R_k=0.30$  and  $R_g=1$ . In general, an increase of  $R_k$  raises the level of the voice fundamental relative to the upper parts of the spectrum. The primary function of the  $R_a$  parameter is to set the frequency  $F_a=1/2\pi T_a=F_0/(2\pi R_a)$  where the source spectrum attains an extra -6dB/oct slope. However, a secondary effect of an  $R_a$  increase - i.e., an  $F_a$  decrease - is a relative boost of the fundamental which occurs in breathy phonation. An increase of  $R_g$ , as

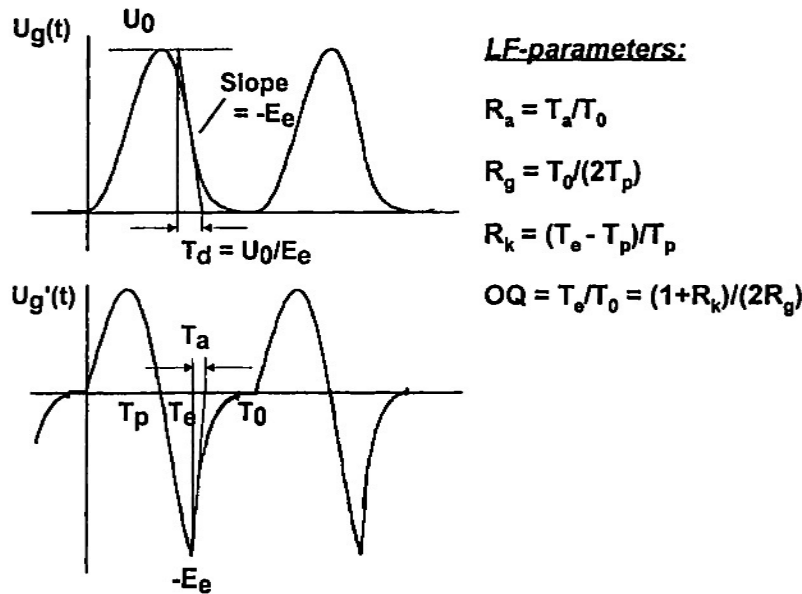


Figure 1. The LF-model extended to include the  $R_d$ -parameter.

in a pressed voice, causes a shortening of the glottal open interval, i.e., a smaller OQ, which promotes the level of the second harmonic at the expense of the fundamental. These relations will be discussed in greater detail in following sections. A sonorous voice has a relatively high  $F_a$  of the order of 2000 Hz.

## 2.1 The $R_d$ -parameter

A statistical and functional analysis of covariation of LF-parameters ranging from an extreme tight, adducted phonation, with low OQ and high  $F_a$ , to a very breathy abducted phonation with high OQ and low  $F_a$  brings out characteristic trends. These can be quantified along a single shape parameter,  $R_d$ , which is closely related to the effective pulse declination time  $T_d = U_o/E_e$  (in ms) of the falling branch, see Figure 1.

$$R_d = (U_o/E_e)(F_0/110) \quad (1)$$

$T_d$  is of the order of 0.5-1 ms for both male and female vowels. The constant 110 in Eq. 1 is dimensionless and was chosen so as to provide numerical equivalence between  $R_d$  and  $T_d$  data of vowels and consonants derived from Gobl (1988) where  $F_0$  averaged 110 Hz. But for a different

scale factor, Eq.1 has recently been adopted by the ICP group in Grenoble.

Within a population of vowels and voiced consonants, we find the following statistical relations (Fant et al., 1994). The added subscript p stands for predicted values

$$R_{ap} = (-1 + 4.8R_d)/100 \quad (2)$$

and

$$R_{kp} = (22.4 + 11.8R_d)/100 \quad (3)$$

An important additional finding is that  $R_d$  can be estimated from the geometrical constraints of the LF model given the measured  $R_a$ ,  $R_k$ , and  $R_g$ .

$$R_d = (1/0.11)(0.5 + 1.2R_k)(R_k/4R_g + R_a) \quad (4)$$

Compared to the definition Eq. 1, this approximation holds with an accuracy of 0.5 dB for  $R_d < 1.4$  and with a maximum error of 1.7 dB at  $R_d = 2.7$  as inferred from Figure 11.  $R_{gp}$  can be derived statistically in the same way as  $R_{ap}$  and  $R_{kp}$  (Fant et al., 1994), but a better approach is to calculate  $R_{gp}$  from  $R_d$  given  $R_{ap}$  and  $R_{kp}$  through Eqs. 2-4. This ensures a conformity with the LF model. The covariation of LF parameters with  $R_d$  is graphically illustrated in Figure 2.



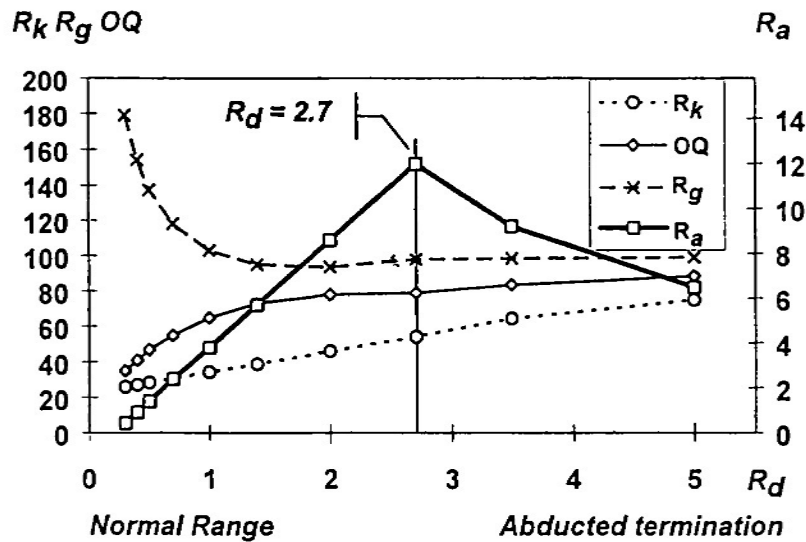


Figure 2. LF-parameters as a function of  $R_d$

The main range of variation is  $0.3 < R_d < 2.7$  whereas the upper range  $R_d > 2.7$  is intended for transitions towards complete abduction as in prepausal voice terminations. A test of the predictability of  $R_a$ ,  $R_k$  and  $R_g$  from  $R_d$  alone is illustrated in Figure 3.

The data pertain to isolated long Swedish vowels produced by a female subject (Karlsson, 1990). From the particular data on  $R_a$ ,  $R_k$  and  $R_g$  values in this article, a single  $R_d$  parameter was computed from Eq. 4 followed by an inversion to derive predicted values  $R_{ap}$ ,  $R_{kp}$  and  $R_{gp}$  through Eq. 2-4, i.e., as a function of  $R_d$  alone. The prediction accuracy happens to be appreciable for this particular female voice. One can also see a correlation between  $R_d$  and  $F_0$ .

The sequence of vowels in Figure 3 are arranged so as to locate unrounded open vowels [æ:] and [a:] in the center and vowels of increasing rounding and tongue constriction, [i:] and [y:] towards the left, and [u:] and [ʊ:] towards the right within the graph. A pronounced vocal tract narrowing, as in the [i:] and [y:] and the maximally rounded [u:] and [ʊ:], causes a loss of transglottal pressure which modifies the glottal flow pulse towards a greater  $R_d$  value and a somewhat lower  $E_c$ . These aerodynamic effects override the pulse skewing (Ananthapadmanabha, 1984; Fant, 1982) associated with increasing supraglottal inductance at constant transglottal pressure. This is also the case for the vowel [o:] in Figure 3 and to some extent also for the [e:] which is rather close to [i:].

In terms of corresponding LF-parameters, the  $R_d$  increase is usually realized as a combination of a greater  $R_k$  and lower  $R_g$  providing a larger open quotient, OQ, which is of the order of 0.7 for [e:], [i:], [y:], [u:], [ɛ:], [o:], and 0.6 for [ɐ:], and 0.55 for [æ:] and [ɑ:]. The  $R_a$  parameter consistently increases with vocal tract narrowing. As a consequence, when  $R_a$  is included in the open quotient  $OQ' = (1 + R_k) / (2R_g + R_a)$ , these contrasts become more apparent. Mean values of the spectral tilt parameter,  $F_a = F_0 / (2\pi R_a)$ , was 375 Hz for [i:], [y:], [u:], [ɛ:], and 990 Hz for [æ:] and [ɑ:]. These findings, although typical of Swedish long stressed vowels, are

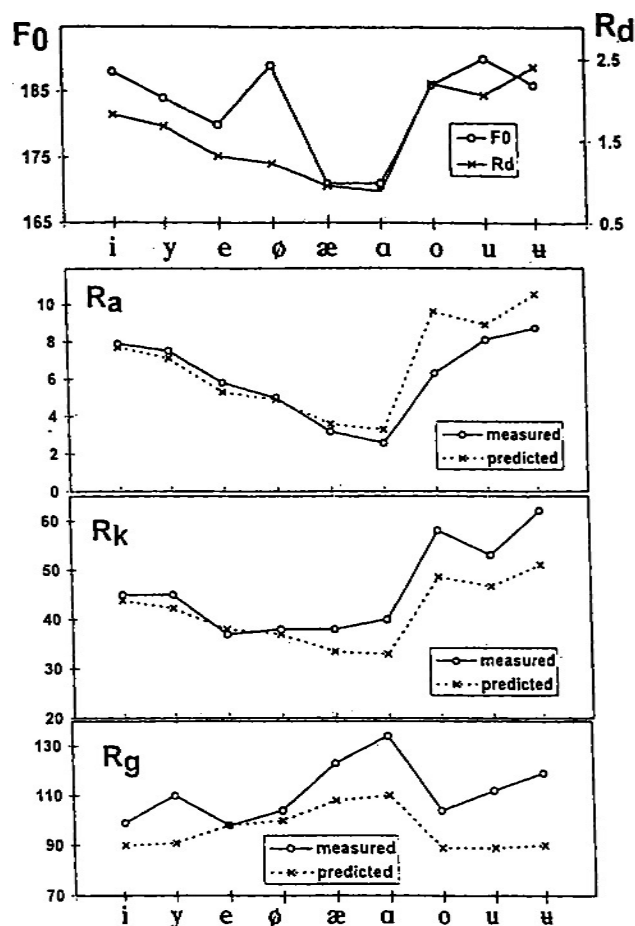


Figure 3. Test of the predictability of LF-parameters from  $R_d$ . Swedish vowels, female speaker.

## THE TRANSFORMED LF-MODEL

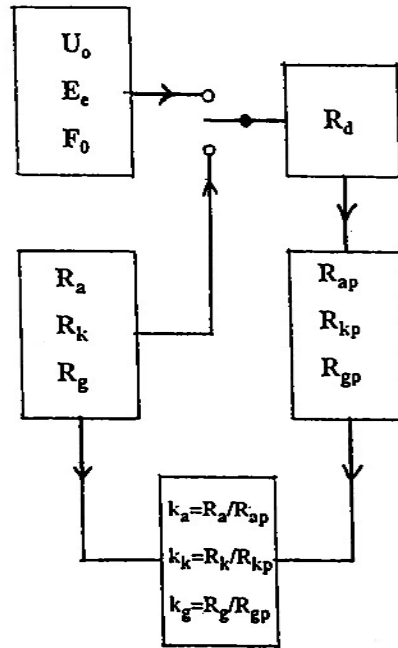


Figure 4. Relations between the original and transformed LF-model.

subject to individual variations with respect to the degree of articulatory narrowing attained in the maximally closed phase, see the discussion in Fant—Kruckenberg (1994).

A conclusion is that essentials of the glottal source wave shape may be contained in a single default parameter,  $R_d$ , defined by Eq. 1, which is relatively easily accessible from a primitive inverse filtering extraction of  $E_e$  and  $U_o$  together with  $F_0$  (Fant—Kruckenberg, 1994) and which has special merits for tracking temporal variations in connected speech. A more detailed analysis requires a full set of measured LF parameters. Deviations of these from default predicted values can be specified in terms of coefficients for

$$k_a = R_a/R_{ap} \quad (5)$$

$$k_g = R_g/R_{gp} \quad (6)$$

$$k_k = R_k / R_{kp} \quad (7)$$

extra aspiration, press, or flow, respectively, as is illustrated in Figure 4, which also indicates the alternative means of calculating  $R_d$  from Eq. 4 given  $R_a$ ,  $R_k$  and  $R_g$ .

Spectra of the transformed LF-model assuming a constant  $E_c$  and  $F_0=100\text{Hz}$  are shown in Figure 5A and 5B. Figure 5A pertains to the glottal flow derivative spectrum and Figure 5B to the double-differentiated spectrum. The sequences of increasing  $R_d$  from 0.3 to 2.7 illustrate the variation from a medially compressed phonation with a small open quotient and a high  $F_a$  to a highly abducted phonation with a large open quotient and a low  $F_a$ . Observe the large variation in the level of the voice fundamental amplitude H1 versus that of the second harmonic H2. This H1-H2 measure, has extensively been used as a voice quality measure, e.g., Klatt—Klatt (1990); Stevens—Hanson (1994); Hanson (1995).

An interesting property of H1-H2 shown in Figure 6 is the almost perfect linear rise with  $R_d$ . In the glottal flow derivative spectrum

$$\text{H1-H2} = -7.6 + 11.1R_d \quad (\text{dB}) \quad (8)$$

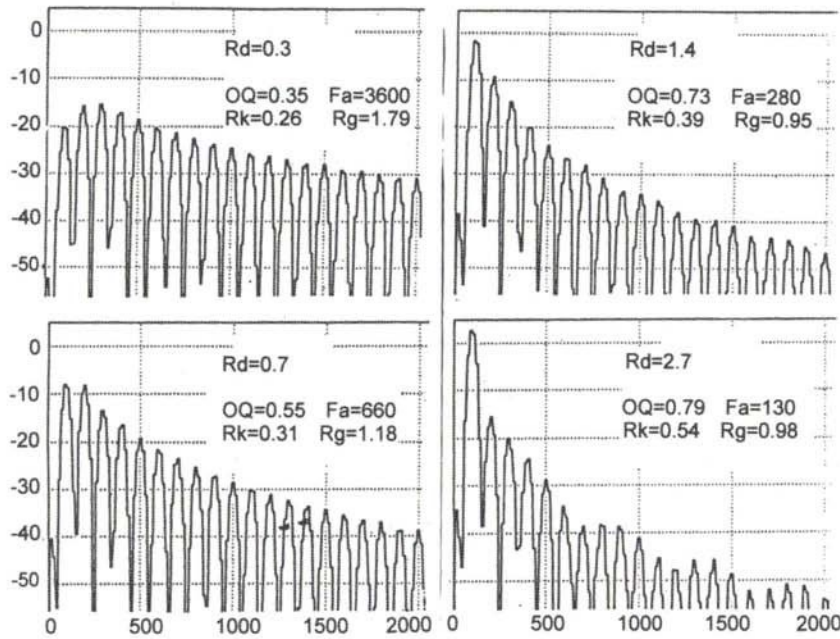


Figure 5A. Glottal flow derivative spectra at  $F_0=100\text{ Hz}$  and varying  $R_d$ .

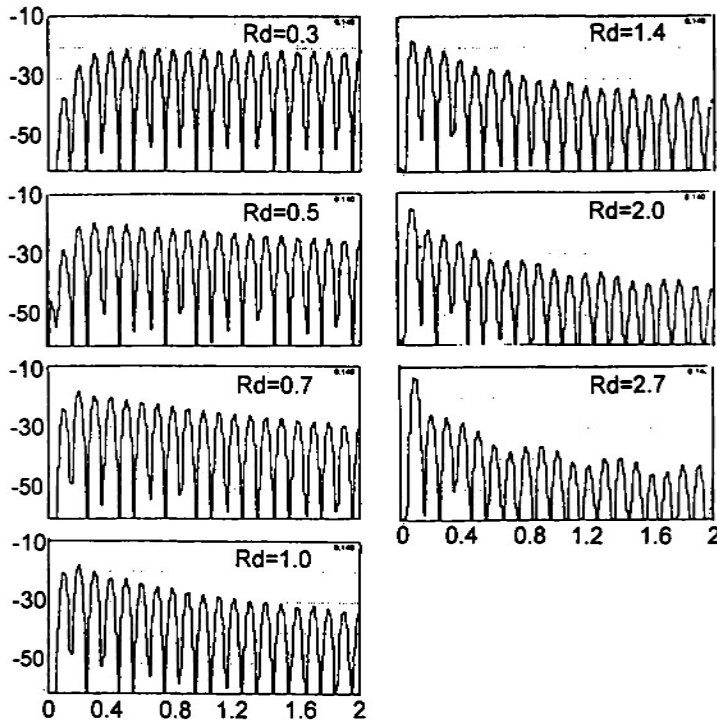


Figure 5B. Glottal flow second derivative spectra at  $F_0=100$  Hz and varying  $R_d$ .

which provides a base for comparison between the two parameter systems. The accuracy is good providing the particular  $R_k$ ,  $R_g$  and  $R_a$  do not depart too much from those predicted from  $R_d$ .

An alternative, see Figure 7, is to refer to the open quotient OQ which requires a more complex regression form.

$$H1-H2 = -6 + 0.27 \exp(5.5OQ) \quad (\text{dB}) \quad (9a)$$

With the alternative definition of the open quotient including the return phase,  $OQ' = (1+R_k)/(2R_g) + R_a$

$$H1-H2 = -7 + 0.715 \exp(3.9OQ') \quad (\text{dB}). \quad (9b).$$

Here OQ is a fraction, not in %. The validity is within 0.5 db in the range of  $0.3 < OQ < 0.7$

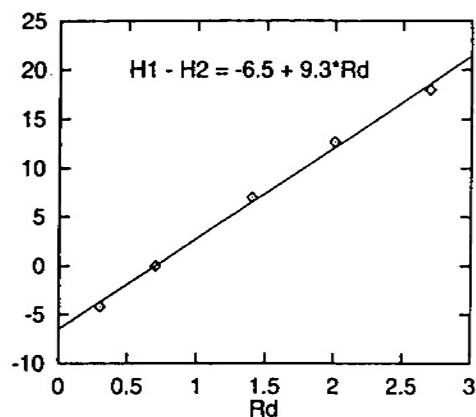


Figure 6. H1-H2 as a function of  $R_d$ .

For the double differentiated spectrum a constant term +6 dB is to be added to Eq. 8 and 9. A higher  $R_d$  and thus higher H1-H2 is typical of female versus male phonations but it is also found in voiced consonants, and aspirated vowels versus regular vowels (Klatt—Klatt, 1990; Stevens—Hanson, 1994). It should be observed that the H1-H2 measure pertains to the source and not the complete sound. In vowels of a high  $F_1$  as [a] and [ae] and low  $F_0$  one can perform an approximate inverse filtering by calculating

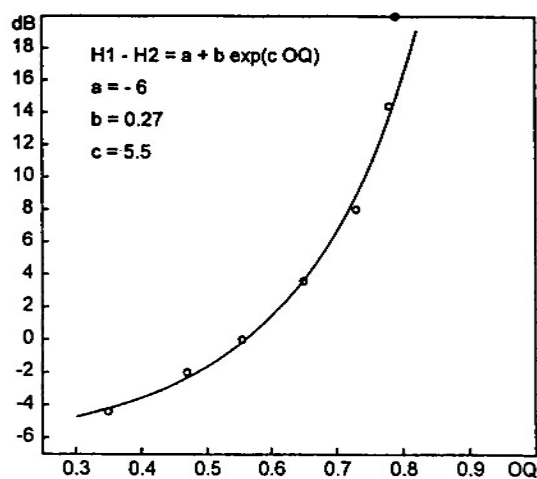


Figure 7. H1-H2 as a function of OQ.

and removing the spectral influence in the low frequency range of  $F_1$  and  $F_2$ , (Stevens—Hanson, 1994). Each of these contributes with a gain of

$$G(f, F_n) = -20 \log_{10}(1 - f^2/F_n^2) \quad (\text{dB}) \quad (10)$$

More exact corrections may be obtained by also considering  $F_3$  and  $F_4$  and higher poles. For sounds with low  $F_1$  in the region of  $F_0$  and  $2F_0$  it is necessary to also consider the bandwidth  $B_1$  which might be difficult to measure.

$$G(f, F_1, B_1) = -10 \log_{10}[(1 - f^2/F_1^2)^2 + (fB_1/F_1^2)^2] \quad (\text{dB}) \quad (11)$$

Accordingly, at  $F_0=100$  Hz and  $F_1=600$  Hz,  $F_2=1000$  Hz as in a vowel [a:] the VT transfer function raises the level of H2 1.2 dB more than H1. The  $F_0$ -dependency is considerable. At  $F_0=200$  Hz and the same F-pattern the correction is 5dB. In female speech the influence of the transfer function is reduced somewhat by higher  $F_1$  and  $F_2$  values but is anyhow more than compensated by the larger H1-H2 of the source spectrum.

### 3. Extended frequency-domain analysis

It is the purpose of the following sections to outline the frequency-domain characteristics of the LF-model and its transformed version in greater detail than in previous publications. This is motivated by the need to handle perceptually important characteristics and to lay a foundation for spectrum matching, i.e., analysis-by-synthesis in the frequency domain, which is a requirement for successful applications in speech synthesis.

#### 3.1 Basic relations

We shall start with an inventory of some basic relations between time-domain and frequency-domain properties. Perceptual attributes are often discussed with reference to relative amplitudes of harmonics and formants derived from FFT. With suitable choice of a window function covering two periods or more a harmonic spectrum may be obtained in which one can observe relative amplitudes, e.g., the fundamental versus second harmonic, H1-H2, as already discussed. However, few people are aware of the precaution needed when studying the temporal variations within a syllable or a phrase of the amplitude of the voice fundamental or of a specific harmonic. The problem is that the FFT and basically DFT do not preserve correct amplitude information when comparing two successive frames of different  $F_0$ . The FFT provides measures of  $G(\omega)/T$  where  $G(\omega)$  is the

Fourier transform and  $T$  the window length. The true harmonic amplitude within a Fourier series, on the other hand, is  $2G(\omega)/T_0 = 2F_0G(\omega)$ . As a result  $F_0$  enters as a hidden scale factor.

The source spectrum may be obtained experimentally by inverse filtering. This process has to rely on a specific definition of the vocal-tract transfer function and on an assumption of linearity. These conditions are fulfilled in essentials only, but not in details such as the presence of interaction ripple and variations within a voice period of the vocal-tract transfer function, i.e., of formant frequencies and bandwidths, including extra poles and zeros. Noise interference may also obscure the separation of source and filter functions. There also appear problems of how to ascribe observable spectral features to the source or to the transfer function. This is largely a problem of coping with uncertainties about the transfer function and pragmatic considerations of defining the source so as to suit a special synthesizer with its inherent constraints and realization of higher pole corrections. This is a problem to be dealt with in the later sections of this paper.

A voice source harmonic spectrum can be decomposed into three components corresponding to three frequency regions:

- (1) The voice fundamental, the spectrum level of which is mainly related to  $U_0$ ;
- (2) The formant domain, the spectrum level of which is mainly related to  $E_e$  and  $F_a$ ;
- (3) A region of spectral prominence around  $F_g = R_g F_0$ . This region is more complex to treat analytically.

The source component of a voiced sound recorded at distance of  $a$  cm from a speakers mouth can be derived from a knowledge of glottal flow data and  $F_0$ , (Fant—Lin, 1988). The amplitude of the voice fundamental is thus

$$A_1 = U_0 k p F_0 (\rho / 4\pi a) \quad (12)$$

The constant  $k$  is of the order of 1 at normal phonations with  $OQ$  around 0.5 but is significantly smaller at high values of  $R_g$  and attains values close to 0.5 for  $R_d = 0.3$  and  $OQ = 0.35$ . The constant  $r$  is the density of air. Given a differentiation of the glottal flow derivative source, which in the frequency domain is executed as a +6dB/oct spectral rise with respect to the voice fundamental, we may express the amplitude of any source harmonic well above  $F_0$  as

$$A''_n = (E_e / \pi) (\rho / 4\pi a) (1 + F^2 / F_a^2)^{-0.5} \quad (13)$$

For frequencies well below  $F_a$  of the LF-model, the double differentiation brings out a flat spectral level  $A_n$ . However, as a reference for voice-source specifications we usually prefer the single differentiation which corresponds to a -6dB/oct slope at high  $F_a$  values. This is motivated by the coherence



with the main trend in a complete vowel spectrum and preserves important relations within the low-frequency domain.

The intermediate region around  $F_g = R_g F_0$  has been referred to as a "glottal formant" (Fant, 1979; Ananthapadmanabha, 1984), which together with the fundamental carries the low-frequency characteristics of a glottal flow pulse. We shall discuss it mainly with respect to the relative amplitude of the second harmonic which is enhanced at high  $R_g$  values. In the absence of a discontinuity, i.e.,  $E_c = 0$ , the limiting value of the flow-derivative spectral slope is -12 dB/oct, i.e., the same as with  $F_a$  approaching zero.

These expressions are discussed in greater detail in (Fant—Lin, 1988) where they were used for an inversion of absolute calibrated spectral data of Swedish vowels from (Fant, 1959) to derive essentials of glottal flow magnitude and waveform.

The main significance of Eqs. 12-13 is that the amplitude of the voice fundamental is proportional to  $U_0$  and formant amplitudes become proportional to  $E_c$ . The ratio

$$(A_1/A''_n) = (U_0 F_0 / E_c) k \pi^2 (1 + F^2 / F_a^2)^{0.5} \quad (14)$$

brings out a factor  $(U_0 F_0 / E_c)$  which we may identify with  $R_d$ , Eq. 1. When  $U_0 / E_c$  is expressed in seconds

$$R_d = 1000 (U_0 F_0 / E_c 110) = U_0 F_0 / (0.11 E_c) \quad (15)$$

Thus,

$$(A_1/A''_n) = 0.11 k \pi^2 R_d (1 + F^2 / F_a^2)^{0.5} \quad (16)$$

For frequencies well below  $F_a$

$$(A_1/A''_n) = 1.1 k R_d \quad (17)$$

which is close to  $R_d$  and suggests a means of determining  $R_d$  from the differentiated source spectrum; compare Eq. 8 where logarithmic values of source harmonic amplitudes are denoted by the symbol  $H$ .

### 3.2 LF-parameter variations

Time and frequency domain properties of the LF-model are illustrated in Figures 8-9.

Figures 8-9 illustrate glottal flow, flow derivative and spectrum of the glottal flow at a voice fundamental period of 10 ms. In all instances, the amplitude value of  $E_c$  has been kept constant. Systematic variations of  $R_a$  in Figure 8 cover the range of  $R_a = 0.4, 0.8, 1.6, 3.2, 6.4$  and  $12.8$  (values in %) corresponding to  $F_a = 4000, 2000, 1000, 500, 250$  and  $125$  Hz with constant

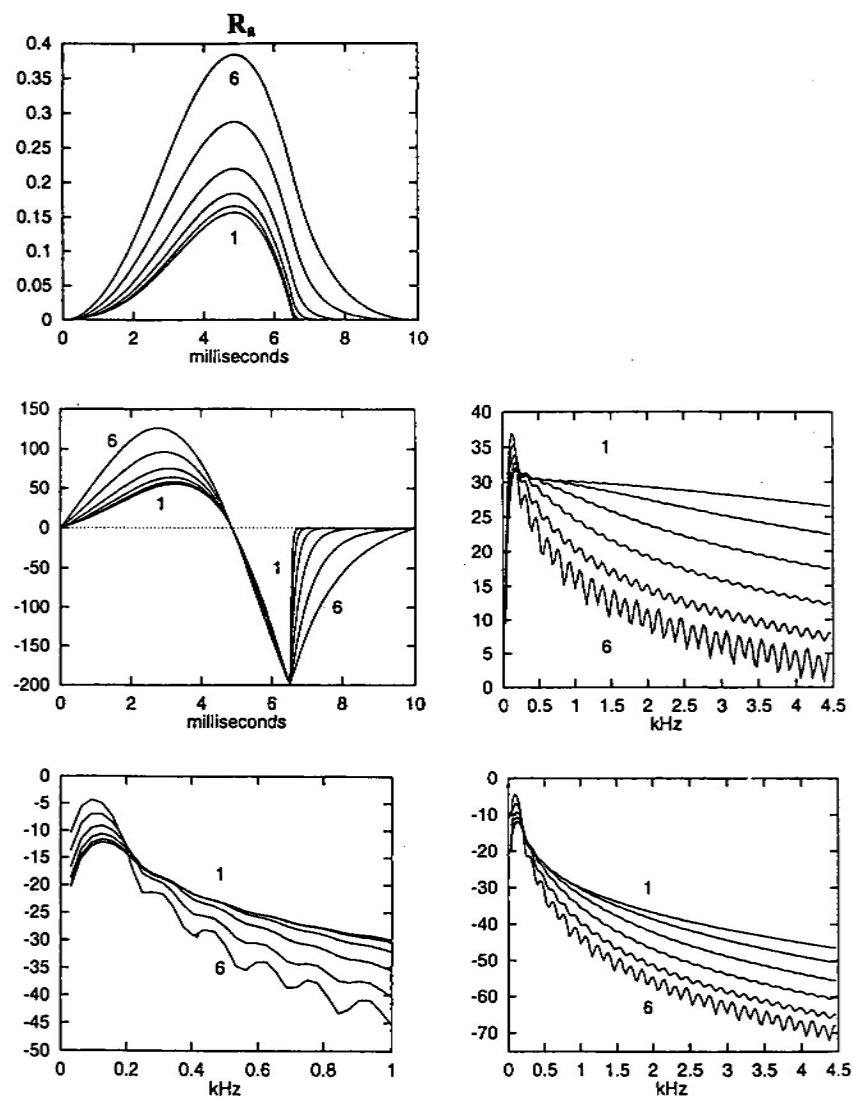
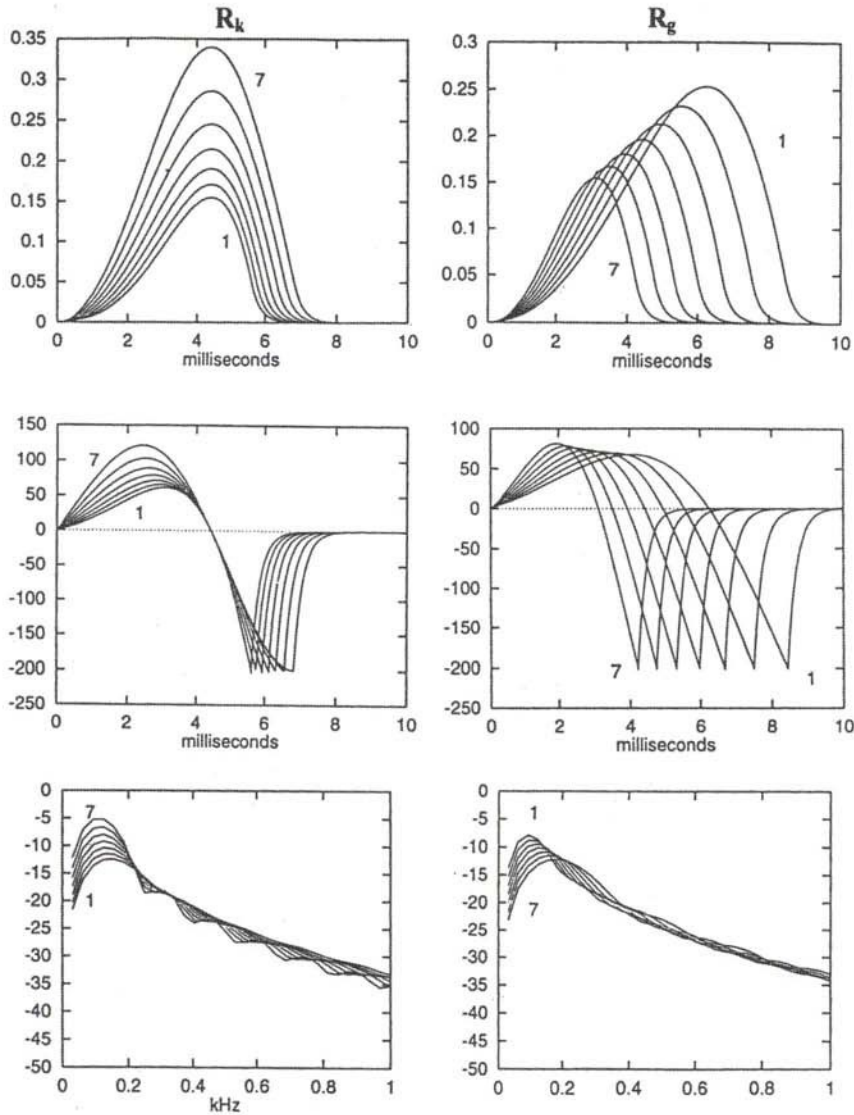


Figure 8. Time and frequency domain properties of the LF-model at constant  $E_c$  with systematic variation of  $R_a$  in six steps labelled 1-6 corresponding to  $F_a=4000, 2000, 1000, 500, 250$  and  $125$  Hz at  $F_0=100$  Hz with constant  $R_g=113$  and  $R_k=38.2$ . The two bottom spectra pertain to the glottal flow derivative, and the upper right spectrum to the double differentiated glottal flow.



**Figure 9.** To the left; time and frequency domain properties of the LF-model at constant  $E_c$  with systematic variation of  $R_k$  in seven steps labelled 1-7 covering the range  $R_k = 27.0, 30.3, 34.0, 38.2, 42.9, 48.2$  and  $54.0$  with constant  $F_a=636$  Hz and  $R_g=113$ . Spectra as in Figure 8. To the right; systematic variation of  $R_g$  in seven steps labelled 1-7 covering the range  $R_g= 80, 90, 101, 113, 127, 142, 160$  with  $F_a=636$  Hz and  $R_k=35$ .

$R_g=113$  and  $R_k=38.2$ . The  $R_k$ -variations (Figure 9), cover the range  $R_k = 27.0, 30.3, 34.0, 38.2, 42.9, 48.2$  and  $54.0$  with constant  $F_a = 636$  Hz and  $R_g=113$ . The  $R_g$  values, also in Figure 9, cover the range  $R_g = 80, 90, 101, 113, 127, 142, 160$  with  $F_a=636$  Hz and  $R_k=35$ . Most of the spectra are limited to the range 0-1000 Hz which focuses on low-frequency contours. For the  $R_a$  parameter in Figure 8, there is included an additional extension to 4.5 kHz and a glottal flow double differentiated spectrum.

Frequency attributes generally referred to are the overall spectrum tilt as set by  $R_a$  and quantified by  $F_a$ , the relative dominance of the fundamental at high  $R_k$  values, and the boost of the second harmonic at high  $R_g$ -values. However, both  $R_k$ ,  $R_a$  and  $R_g$  show a range of about 7dB variation in the region of the voice fundamental.

As a means of facilitating LF-parameter extraction by spectrum matching the following differential data has been assembled.

*Table 1.* Change in each of  $R_a$ ,  $R_k$  and  $R_g$  needed to increase the level of the fundamental H1 by 1 dB and H1-H2 by 1 dB keeping other parameters constant.

| Parameter | dR/dH1 | dR/d(H1-H2) |
|-----------|--------|-------------|
| $R_a$     | 1.0    | 1.25        |
| $R_k$     | 3.0    | 4.5         |
| $R_g$     | -12    | -10         |

In Figure 6, we noted a linear relation  $H1-H2 = -7.6 + 11.1R_d$  which assumes default values of  $R_a$ ,  $R_k$  and  $R_g$ , i.e., those predicted from Eq. 2-4. To what extent does this relation hold for other combinations of  $R_a$ ,  $R_k$  and  $R_g$  satisfying the same  $R_d$ ? Assuming a factor  $k_a=2$  increase of  $R_a$ , i.e., a halving of  $F_a$ , at constant  $R_g=118$  and  $R_d = 0.7$ , which implies an  $R_a$  change from 2.36 to 4.72%, the H1-H2 increases by  $2.36/1.25=1.9$  dB which is partially compensated by a decrease of 1.5 dB due to the change in  $R_k$  from 30.7 to 24 required to maintain constant  $R_d=0.7$ . On the other hand, a change in  $R_g$  from 118 to 100 and a change in  $R_k$  from 30.7 to 27.5 at constant  $R_d$  changes H1-H2 by  $(1.8-0.7)= 1.1$  dB. Here the  $R_g$  variation dominates the shift in H1-H2, Eq. 8-9.

An alternative use of Table 1 is to determine combinations of LF-parameters that provide invariant values of H1 or of H1-H2. It is thus directly seen that an increase of  $R_g$  by 10 percent causes a decrease of H1-H2 by 1 dB which is fully compensated by 4.5 percent increase in  $R_k$  or by 1.25% increase in  $R_a$ . At the same time, H1 is increased by 0.5 dB respectively 0.4 dB. The simultaneous increase of  $R_g$  and  $R_k$  above causes a

change in  $R_d$  from 0.7 to 0.77 and a minor change in  $OQ$ , from 0.55 to 0.53. The main change in the time domain is a delay of the onset of the glottal flow pulse which shortens the rise time and increases the fall time and thus causes a less forward skewed shape. The rise in  $H1$  and  $H2$  is seen as a relative lowering of  $H3$ .

### 3.3 The $R_d$ -parameter

A systematic variation of  $R_d$  is illustrated in Figures 10-12 which show glottal flow and flow-derivative wave shapes and flow-derivative spectra. The corresponding default predicted LF parameters are listed in Table 2.

Figure 10 was constructed keeping  $E_e$  constant during the  $R_d$  variation. This implies almost 20 dB increase of  $U_o$  from  $R_d=0.3$  to  $R_d=2.7$ .

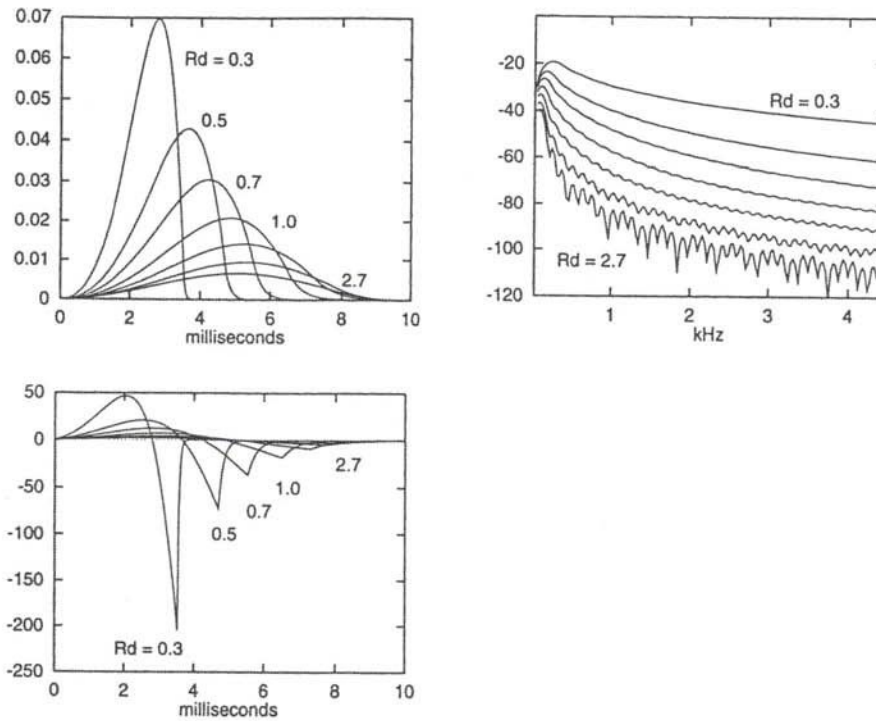


Figure 10. Glottal flow, flow derivative and flow derivative spectrum of the transformed model at constant  $E_e$  with  $R_d = 0.3, 0.5, 0.7, 1.0, 1.4, 2.0, 2.7$ .

In Figure 11,  $U_0$  is nominally held constant with  $E_c$  covering a 20 dB range. In practice, since  $U_0$  is derived from  $R_d$  of the approximate formula Eq.4, the peak values of the glottal flow at  $R_d$  higher than 2 come out

Table 2. Default LF-parameters for a range of  $R_d$  values.

$F_a = 1/2\pi R_a$  refers to  $F_0 = 100$  Hz

| $R_d$ | $R_a$ | $F_a$ | $R_k$ | $R_g$ | OQ   |
|-------|-------|-------|-------|-------|------|
|       | %     | Hz    | %     | %     | %    |
| 0.3   | 0.44  | 3600  | 26.0  | 179.0 | 35.0 |
| 0.5   | 1.40  | 1590  | 28.3  | 137.0 | 47.0 |
| 0.7   | 2.36  | 674   | 30.7  | 118.0 | 55.5 |
| 1.0   | 3.80  | 420   | 34.2  | 103.0 | 65.0 |
| 1.4   | 5.70  | 280   | 39.0  | 95.0  | 73.0 |
| 2.0   | 8.60  | 185   | 46.0  | 93.5  | 78.0 |
| 2.7   | 12.00 | 133   | 54.3  | 98.0  | 79.0 |

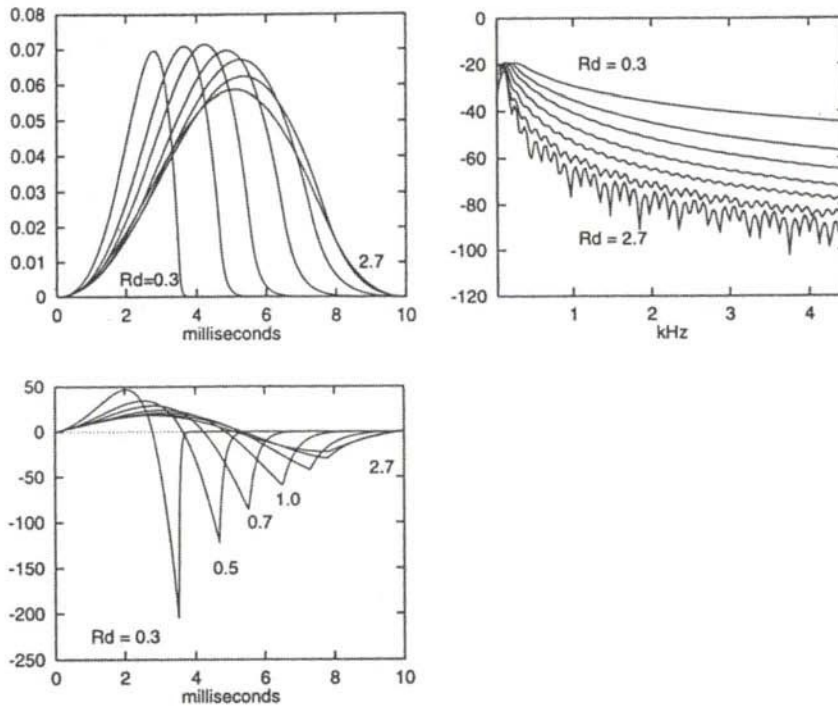


Figure 11. Time- and frequency domain properties of the transformed model with approximately constant  $U_0$  and varying  $E_c$

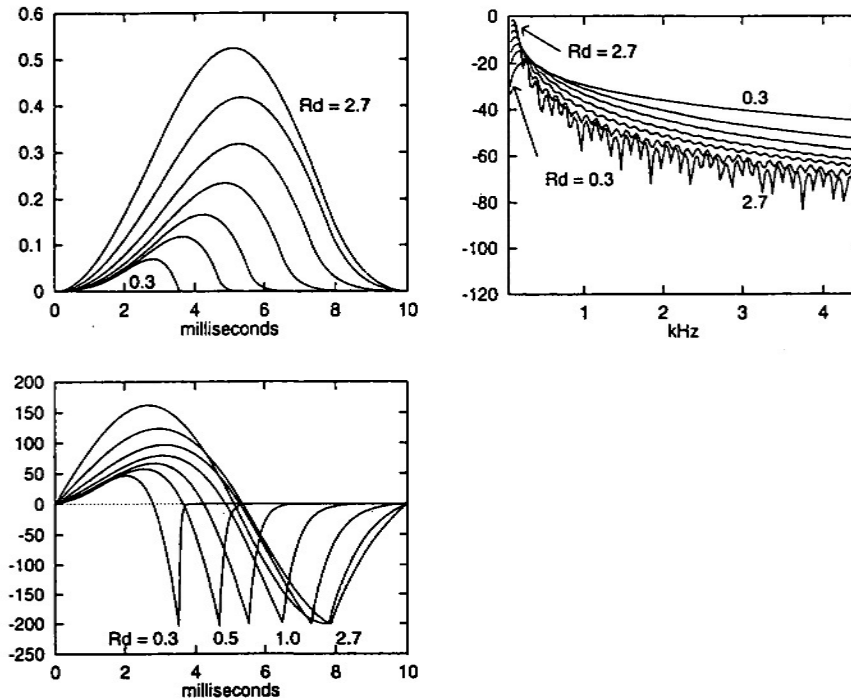


Figure 12. Time- and frequency domain properties of the transformed model with  $U_0$ -variations scaled one half of  $E_c$  variations on a dB-scale.

somewhat smaller than ideally. The error is maximally 1.7 dB at  $R_d=2.7$  which we shall ignore in the following treatment.

In connected speech we seldom have a situation of constant  $E_c$  and varying  $U_0$  but more often varying  $E_c$  at constant  $U_0$ , e.g. at the end of a breath group. A more common relation is that of Fig 12 where, in the range from  $R_d=0.3$  to  $R_d=2.7$ ,  $E_c$  has been decreased by 40 dB and  $U_0$  by 20 dB. A fourth possibility is that  $R_d$  is constant while  $E_c$  and  $U_0$  vary proportionally.

As a consequence of the assumptions above, the glottal flow derivative spectra show a high frequency range expanding from Figure 10 to Figure 12. The low frequency region is almost constant in Figure 11, where  $U_0$  is constant, but is dominated by high  $R_d$  values in Figure 10 and by low  $R_d$  values in Figure 12

## 4. Inverse filtering and spectrum matching

### 4.1 Ambiguities in LF-parameter extraction

Conventional LF-parameter extraction from inverse filtering is far from an exact procedure, and is subject to variations in individual strategies, and thus, runs the risk of being more an art than a science.

Figure 13 illustrates a difficult case. It pertains to a vowel [i:] showing glottal flow derivative waveform and spectrum after inverse filtering with  $m=7$  cancellation zeros above and  $m=8$  zeros below. Sampling frequency was 16 kHz. The analysis was done by IK, our expert on LF-parameter extraction. The left hand parts pertain to time-domain matches of the return phase and the right hand parts to spectral matching in the 3000 Hz domain. For  $m=7$  zeros, the time domain match gave  $F_a=424$  Hz, while the spectral match resulted in a much higher value,  $F_a=1700$  Hz. With  $m=8$ , these values were lower,  $F_a=340$  and 890 Hz, respectively. An alternative setting with  $m=7$  to match the 3500–4000 Hz region, which is the perceptual mean of the upper part of the spectrum, gave  $F_a=5600$  Hz. Depending on the number of cancellation zeros  $m$ , their particular positions and matching criteria,  $F_a$  accordingly varied between the extreme limits of 340 and 5600 Hz.

An additional difficulty is apparent in the temporal contour of the glottal flow derivative waveform which exhibits a prominent double-peak profile typical of interaction (Fant, 1986; Lin, 1990; Båvegård—Fant, 1995). Here is an apparent ambiguity in the location of the starting point of glottal opening. The investigator choose to ignore the first peak which implies greater  $R_k$  and  $R_g$  values than if the starting point had been set somewhat earlier. The uncertainty in the location of the starting point is a recurrent problem in LF-parameter extraction.

This is, however, a rather extreme example of difficulties. Experienced investigators using the same strategy generally differ less than 10 % in their measurements.

### 4.2 Analysis by synthesis. Spectrum matching.

Frequency-domain matching is frequently used as a complement to the basic time-domain inverse filtering to derive LF-parameters (Karlsson, 1992). A more radical frequency-domain method is analysis-by-synthesis varying the control parameters of a synthesizer to provide a best match. The latter method has the merit of securing an optimal representation for practical use but may require substantially different parameter settings than derived from conventional inverse filtering. A major problem is the nonuniqueness of



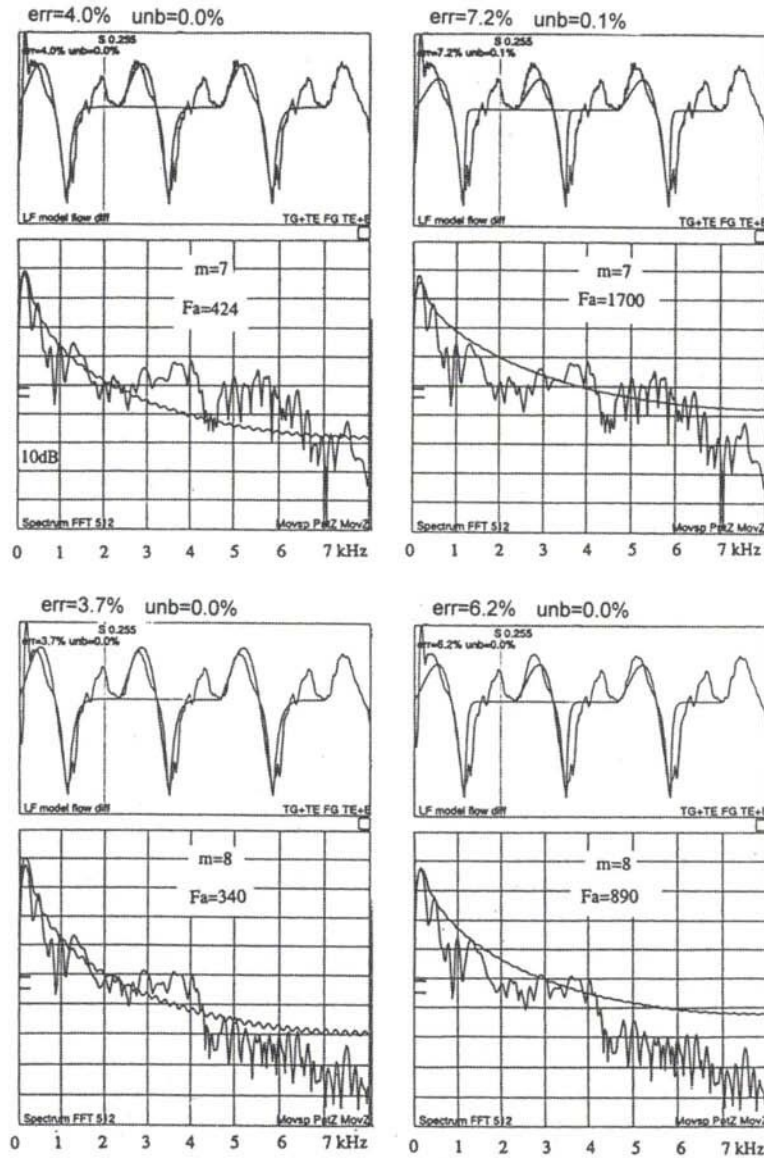


Figure 13. Variability in  $F_a$  determinations from inverse filtering of a vowel [i:]. The upper two graphs pertain to the use of  $m=7$  zeros and the lower two graphs to  $m=8$ . The left hand graphs pertain to time-domain fitting and the right hand graphs to frequency domain fitting.

source-filter decomposition and the pragmatic need of establishing synthesis routines (Fant, 1995). The selection of source parameters will depend on the type of synthesizer, the inventory of digital pole and zero circuits employed, the sampling frequency and the particular setting of formants and antiformants.

As a rule of thumb, cascade formant synthesis of all-pole sounds should not need any special higher-pole correction if poles are evenly distributed up to half the sampling frequency,  $F_s/2$ . However, the number of poles,  $m$ , is crucial and should be related to the length of the speakers vocal tract. As shown in Fant (1955) the condition is

$$m = F_s(l_e/c) \quad (18)$$

Thus,  $m=8$  poles is the exact requirement for the standard  $l_e=17.65$  cm of a male vocal tract. A number of  $m=9$  poles would be required for  $l_e=20$  cm which would be the average for a large male tract or the [u:] configuration for a standard male. Female and child vocal-tract lengths vary from 16 to 13 cm. In specific, a number of  $m=7$  poles would suit a female with  $l_e=(7/8)17.65=15.5$  cm, i.e., a formant scale factor of  $8/7=1.14$  and  $m=6$  would correspond to  $l_e=13.2$  cm and a formant scale factor of  $8/6=1.33$ .

These rules are important to consider in conventional inverse filtering as well as in analysis by spectral synthesis. The overall VT length also varies with the particular articulation.

Too many poles in female synthesis will provide an excessive high level in the  $F_4$ - $F_5$  region and too few poles in male synthesis will have the opposite effect. There is apparently a trading relation between  $F_a$  and the number and setting of formants above  $F_3$ , but as a rule an optimal  $F_a$  setting can not fully compensate a non-representative number of formants.

The general relations between LF-parameters and spectral detail outlined in the previous chapter have found good use in our spectrum matching experiments with the GLOVE synthesizer (Karlsson—Neovius, 1993) which has a standard set of  $m=7$  digital formant circuits in cascade in the vocalic branch. The procedure involved a narrow-band sectioning of the vowel to produce a harmonic spectrum. The exact  $F_0$  was noted and applied to the synthesizer. In order to secure a good view of the low frequency region most comparisons were made in the glottal flow derivative domain and not in the second derivative domain.

As shown in Figure 14 we managed to obtain a quite good match of the spectrum of an [ɑ:] up to 3500 Hz and an [i:] up to about 4000Hz. The [ɑ:] required an  $F_a$  of 4000 Hz and [i:] a somewhat lower  $F_a$  which agrees with the inverse filtering data. The [i:], earlier discussed in connection with Figure 15, shows traces of an intruding zero at 3100 Hz and a pole at 3000

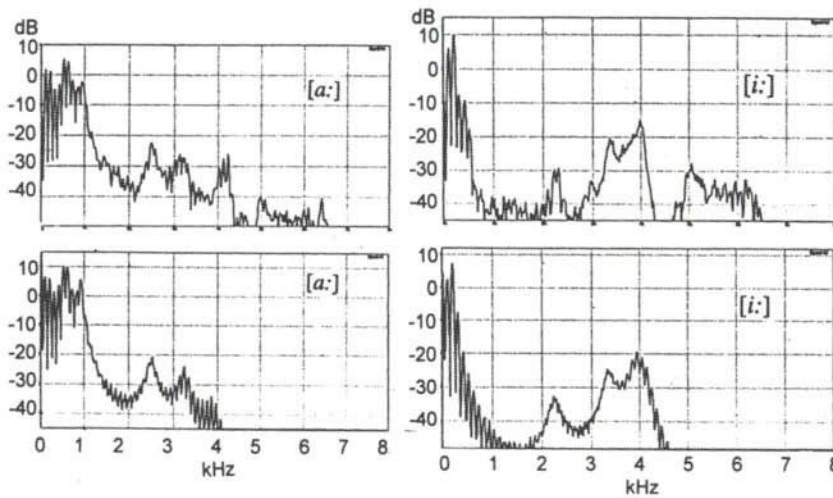


Figure 14. Analysis-by-synthesis spectrum matching of a vowel [a:] and a vowel [i:] with  $m=7$  formants and  $F_3=16$  kHz. Originals above and synthesis below.

Hz of unknown origin. The spectral peak at 3300 Hz is identified with  $F_3$ . The [i:] also shows a spectrum reinforcement at 400 Hz 10 dB below the  $F_1$  level which probably reflects interaction ripple (Fant, 1986, 1993; Båvegård—Fant, 1995), see Figure 13.

Both [a:] and [i:] showed spectral energy above 4000 Hz which is not maintained in the synthesis. In specific, the prominent  $F_4$  at 4100 Hz in [a:] is missing in the synthesis. This high-frequency region combined with a spectral minimum at 4500 Hz is often seen in male voices. It appears to be related to the combined filtering of the larynx tube shunted by the Piriformis cavities.

Adding an 8th formant circuit brings in the spectral peak just above 4000 Hz in [a:] as shown in Figure 15. At the same time we could, as expected, reduce the value of  $F_a$  from 4000 Hz to 1500 Hz. In the low frequency range, the relations between the first two harmonics,  $H_1$  and  $H_2$ , and the complete set of LF-parameters outlined in Table 1 served as an efficient means for iterative search for a best match.

Figure 16 exemplifies variations which pertain to a stressed and an unstressed modification of the [a:]. For simplicity we have chosen the same  $E_c$  and  $U_o$  and  $F_0$  and thus the same  $R_d=0.7$  but different combinations of  $R_k$ ,  $R_g$  and  $F_a$ . The unstressed version has the default values of  $R_k=31$ ,

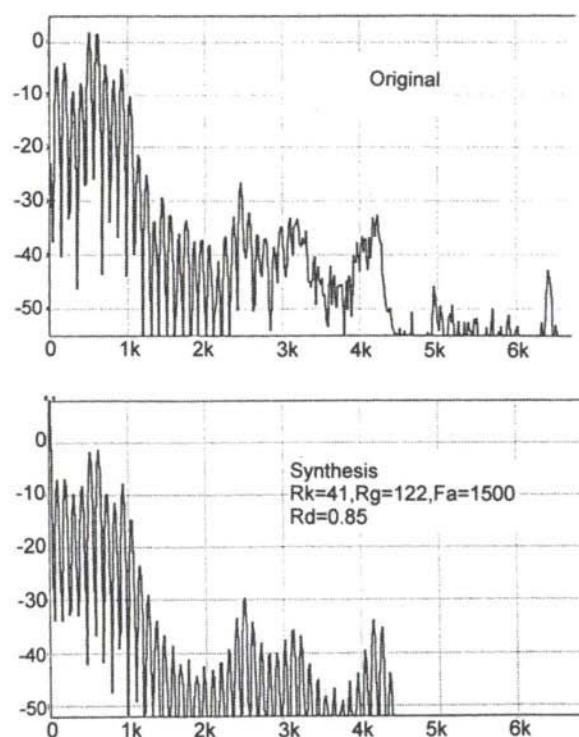


Figure 15. Original and synthesis of a vowel [ɑ:] with  $m=8$  formants

$R_g=118$  and  $F_a=680$ . The stressed version has the values  $R_k=47$ ,  $R_g=177$  and  $F_a=2700$  and also a smaller open quotient,  $OQ=0.4$  compared to  $OQ=0.55$  for the unstressed case. In terms of LF coefficients, Eq. 5-7, the stressed version accordingly has a  $k_a=680/2700=0.25$  and a  $k_g=177/118=1.5$ . The greater efficiency of the stressed phonation is associated with 2 dB greater overall SPL and a 6 dB higher level of F3.

which is not an uncommon finding, see e.g., Sluijter (1995). There is also a 2 dB greater prominence in H2 and H3 which adds to the sound quality. This kind of laryngeal stress adjustment does not require any raise in the subglottal pressure and the air consumption is actually somewhat less. It is often found in phrase-final moderate stress. In contrastive and emphatic stress the differences are greater and there is an decrease of  $R_d$  with stress.

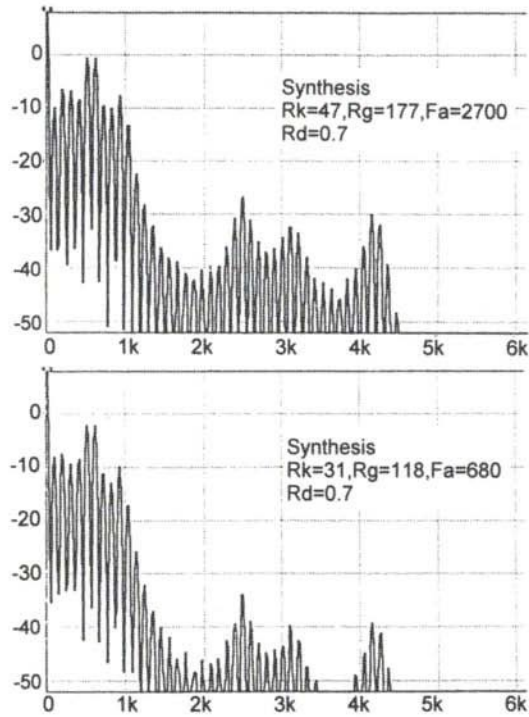


Figure 16. Synthesis of a vowel [ɑ:],  $m=8$ , with two different combinations of LF-parameters at constant  $E_c$  and  $R_d=0.7$ , below default values, above with increased  $F_a$  and  $R_g$  ( $k_a=0.25$ ,  $k_g=1.5$ ) and also increased  $R_k$  to maintain constant  $R_d$ . These could simulate unstressed and stressed phonation.

### 5. Spectral shifts at varying voice intensity

Figure 12, already referred to, has a special relevance for modelling source-spectrum changes at varying voice intensity but also for the main trends of covariation of excitation strength and waveform in connected speech. The common denominator is that the  $U_o$  range, and thus the  $R_d$  range, is about one half of the  $E_c$  range.



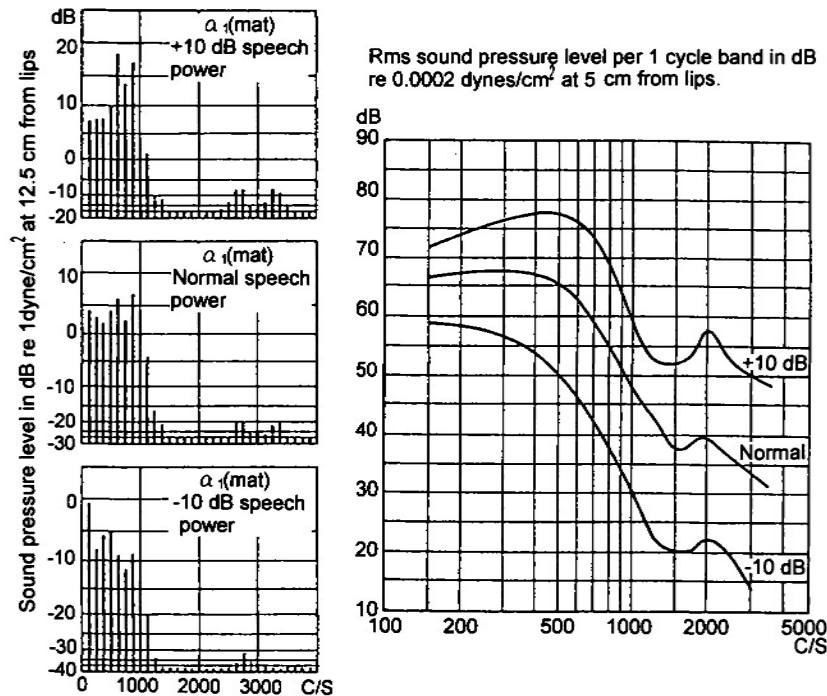


Figure 17. Spectral variation at varying voice intensity. (From Fant, 1959, 1980).

The nonlinear spectral shifts at varying voice effort illustrated in Figure 17 derive from an earlier study (Fant, 1959, 1980). It shows the spectrum of a vowel [a:] and the long-time average spectrum of a sentence produced with normal voice effort and with nominally 10 dB higher and 10 dB lower voice intensity as monitored by a sound-pressure level meter. These curves show the typical pattern of the high-frequency gain increasing at a higher rate than the low-frequency gain at increasing voice effort. Within an additional study of nine sustained Swedish vowels the average trend was that 10 dB increase of the level of F1 was accompanied by 4 dB increase in the level of the fundamental, 14 dB in F2, 16 dB in F3, and 14 dB in F4.

The question is now if these shifts can be interpreted in the frame of the transformed LF-model. As a test material the long-time average spectrum curves were selected.

As shown in Figure 18, a perfect fit (within 0.5 dB) between the high (+10 dB) minus the low voice effort (-10 dB) spectral levels at the fundamental 125 Hz and at 500 Hz, 1500 Hz and 2500 Hz was achieved,

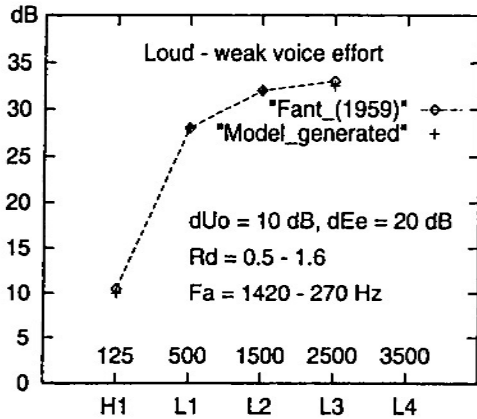


Figure 18. Spectral difference between high and low voice intensity from Figure 17, matched by the transformed LF-model.

assuming that  $U_o$  was reduced by 10 dB and  $E_e$  by 20 dB in the range of  $R_d = (U_o/E_e)(F_0/110)$  varying from 0.5 to 1.6 and with  $F_a$  values determined from  $R_d$ .

The spectral level shifts of 10 dB at 125 Hz, 27.5 dB at 500 Hz, 32 dB at 1500 Hz, and 2500 Hz at increased voice effort can accordingly be decomposed in three components:

- (1) An increase of  $E_e$  by 20 dB affecting formant regions
- (2) An increase of  $U_o$  of 10 dB affecting the fundamental
- (3) An increase of  $F_a$  from 275 to 1420 Hz providing a high-frequency boost, Eq. 10, which amounts to 13 dB at 2500 Hz

The particular range of  $R_d$  from 1.6 to 0.5 is optimal but not crucial. Fairly good fits are obtained with other 10 dB ranges. These general trends of the spectrum level in the  $F_0$  region varying much less than spectral levels at higher frequencies were also observed by Gauffin—Sundberg (1980) in a study of a vowel [æ] sustained at various voice intensities by singers and nonsingers. The only significant difference between these two groups was that singers produced a relatively larger difference in the 2900 Hz band (singer's formant?).

These trends, on a smaller scale, have also been found in connected speech comparing parts of phrases spoken with different voice levels or comparing focally stressed syllables with unstressed syllables (Fant—Kruckenberg, 1994). A stress scale has been constructed on the basis of a scale unit of one tenth of the range above, i.e., 2 dB in  $E_e$  associated with 1 dB in  $R_d$  and in the voice fundamental, and a 1.2 dB extra high-frequency

boost. A relatively prominent contrastive focal stress would require a shift of about 4 such units.

## 6. Bandwidths

Breathy voicing and transitional regions affected by aspiration involve higher  $R_d$  values, lower  $E_e$ , traces of subglottal poles and zeros, noise and significant bandwidth increase (Fant—Lin, 1988; Klatt—Klatt, 1990; Karlsson—Neovius, 1993; Fant—Kruckenberg, 1995). Calculations have recently been undertaken to estimate the bandwidth increase due to losses in the kinetic and the frictional glottal resistance in combination with the glottal inductance and a shortcircuited subglottal tract. Bandwidth increase of the first and second formants of [a] and [e],  $DB_1$  and  $DB_2$ , as a function of the glottal area  $A_{g0}$  including a possible posterior bypass, are shown in Figure 19.

The derivation is based on tabulated data of purely resistive loads (from Fant, 1960), corrected for the influence of the glottal inductance. A transglottal pressure drop of 4 cm H<sub>2</sub>O was assumed. Higher values did not cause radical reductions. The main trends conforming with experience from

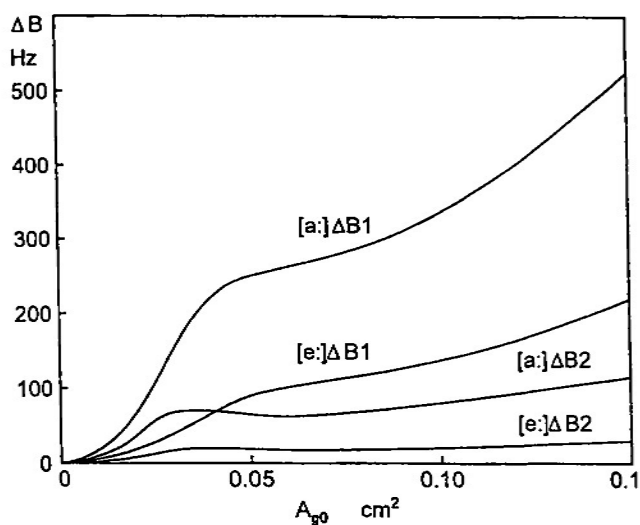


Figure 19. Calculated increase of first and second formant bandwidths of [a] and [e] taking into account glottal kinetic and frictional resistance and a finite glottal inductance.



speech analysis are the greater influence on  $DB_1$  in [a] than in [e] and that  $DB_1$  is always considerably larger than  $DB_2$ . It is of interest to note that significant effects occur already at  $A_{g0}=0.02 \text{ cm}^2$  and that the values are rather constant between  $A_{g0}=0.05$  and  $0.1 \text{ cm}^2$ . These data should be checked with the subglottal tract included.

As a rough estimate of the covariation of bandwidth increase with formant frequencies and the  $R_a$ -parameter the following formula are suggested.

$$DB_1 = 250(F_1/500)^2(R_a/12) \quad (19)$$

$$DB_2 = (DB_1/2)(F_1/F_2) \quad (20)$$

## 7. General conclusions

The LF model and its transformed version have been described in greater detail than in previous presentations. The normal covariation among LF-parameters and the basic waveshape parameter  $R_d$  permits the prediction of default LF parameters from  $R_d$  which is a unique function of  $U_0$ ,  $E_c$  and  $F_0$  and can be derived from these with simplified inverse filtering techniques, (Fant, 1993; Fant et al., 1994; Fant—Kruckenberg, 1994). Conversely, given the complete set of LF-parameters a unique  $R_d$  value can be calculated, Eq. 4.

The main importance of the  $R_d$  parameters is that it is the most effective single measure for describing voice qualities and that it simplifies the description of text-to-speech source rules. The main range is  $0.4 < R_d < 2.7$  whereas the domain  $R_d > 2.7$  is essentially devoted to transitions into abducted termination of voicing. A low  $R_d$  is associated with effective glottal closure and small opening quotients set by relative large  $R_g$  values, whereas high values of  $R_d$  are typical of abducted phonation and high values of open quotients. In general, male voices have smaller  $R_d$  than female voices but there is an apparent overlap, and voiced consonants have larger  $R_d$  than the speakers vowels. Voiced consonants generally show larger  $R_a$ , i.e., lower  $F_a$  values than predicted from  $R_d$  alone. This is also typical of aspirated vowels and is expressed as an aspiration factor  $k_a = R_a/R_{ap}$  greater than 1. Here we find an exception from the general rule of  $R_k$  increasing with  $R_a$ .

The spectral domain properties of the LF-model and its transformed version have been analyzed in greater detail than in previous studies. Of specific interest is the finite contributions of all LF-parameters to the low

frequency spectral shape. The relative level of the amplitude of the voice fundamental H1 is thus affected not only by  $R_k$  but also by  $R_g$  and by  $R_a$  (or  $F_a$ ). A linear relation is found between H1-H2 and  $R_d$ , Eq. 8. The exponential relation between H1-H2 and the open quotient OQ, Eq. 9, is more invariant with respect to combinations of LF parameters.

Special attention has been devoted to the covariation of excitation amplitude  $E_e$  and  $R_d$  or  $U_o$ . A typical pattern is that in the log amplitude domain  $R_d$  and  $U_o$  variations tend to be one half of the corresponding  $E_e$  variations. Earlier data on spectral changes with increasing voice effort support this trend. Thus, increasing  $U_o$  with 10 dB raises the amplitude level of the voice fundamental, H1, by approximately the same amount, whereas a simultaneous rise in  $E_e$  of 20 dB acts as a scale factor in the formant range to which adds 10-15 dB due to lower  $R_d$  and thus higher  $F_a$  values. This represents a relatively large range of vocal effort. Similar trends, but of a lesser magnitude, are found to accompany marked instances of stress in connected speech, presumably connected with an increase of pulmonary pressure. However, increased intensity or increased glottal excitation are not necessary attributes of stress (Fant—Kruckenberg, 1994, 1995; Stevens, 1994; Sluijter, 1995). We have fresh evidence of instances where stress is realized by increased  $F_a$ , decreased  $R_d$  and also by a general shortening of the glottal pulse through increase of the  $R_g$ -parameter. This is related to a more distinct onset of the glottal open phase. In the frequency domain there is a resulting decrease of the H1-H2 measure.

Conventional methods of LF-parameter extraction are subject to many sources of uncertainty. Alternative determinations of  $F_a$  from time and frequency-domain fitting may show very large variations depending on the particular frequency region aimed at and also on the particular number of inverse filter zeros, their locations and the possible addition of extra pole-zero pairs. However, with similar strategies and conventions trained investigators usually agree within 5-20 % in  $R_k$ ,  $R_g$  and  $R_a$  determinations.

For synthesis applications many of the ambiguities involved in inverse filtering may be avoided by searching for the set of LF-parameters that provide a best match in combination with a particular synthesizer and its constraints. This technique is optimal for HiFi copy synthesis. Special attention has to be devoted to the role of the additional high-frequency gain associated with digital realization of formant circuits which can be considered as a part of a higher pole correction. It is thus found that  $m=8$  or 9 poles up to  $F_s/2 = 8$  kHz is needed for male synthesis and  $m=6-7$  poles for child or female synthesis. A general conclusion is that the number of formant circuits should be  $m=F_s(l_e/c)$ , where  $l_e$  is the effective length of the vocal tract. The normal variations of  $l_e$  with articulation, i.e., with the

particular vowel, should also be considered. Our findings have provided a theoretical support for earlier expressed needs of nine zeros in inverse filtering of male voices (Gobl, 1988) and the practice of  $m=9$  formants for synthesis as a reference for speech recognition (Blomberg, 1994).

Matching a low frequency region requires manipulation of all LF parameters while the upper part of the spectrum is mainly determined by  $F_a$ . A systematic use of the relations between minimal variations in LF parameters and corresponding changes in spectral shape parameters such H1 and H1-H2 and formant levels simplify the matching procedure. Analysis-by-synthesis performed by spectrum matching according to these principles have been quite successful. As an example, the amplitudes of the 10 first harmonics of the vowel [a] were reconstructed with mean errors of the order of 1 dB which also applied to the spectral levels of F3 and F4.

Glottal source and vocal tract dynamics are intimately connected and influenced by prosody. These problems are mainly dealt with in other studies (Fant—Kruckenberg, 1994, 1995) and have only briefly been referred to in the text. Consonant-vowel source contrasts are enhanced by stress (Fant—Kruckenberg, 1994, Gobl, 1988). The greater articulatory prominence causes a more efficient vocal-tract closure which affects the source characteristics of voiced consonants. The tendency of increased  $R_d$  and OQ and thereby decreased  $F_a$  is similar to what appears in aspiration. A common denominator appears to be the state of vocal-fold abduction which in voiced consonants is caused by the reduced transglottal pressure drop and the increased supraglottal air pressure. This aerodynamic effect overrides a potential  $E_e$  increase associated with increased supraglottal inductance causing pulse skewing at constant transglottal pressure drop.

In Swedish, not only voiced consonants but also narrow vowels, are articulated more narrowly under the influence of stress which counteracts the normal voice source correlates of prominence, i.e., the increased  $E_e$  and lower  $R_d$  and higher  $F_a$ . In general, source rules for connected speech thus have to be structured relationally. Some data for a few consonants and several speakers support this view. Consonant data have to be related to associated vowels and to the degree of articulatory emphasis which determines the degree of consonant-vowel contrast. The Swedish maximally close vowels such as [i:] and [u:] when stressed are realized with a diphthongal movement towards a homorganic voiced consonant, the intensity of which is negatively correlated to stress (Fant—Kruckenberg, 1994).

The bandwidth broadening associated with glottal abduction is an important component of aspiration correlates (Fant—Lin, 1988; Klatt—Klatt, 1990; Karlson—Neovius, 1993; Gobl—Ni Chasaide, 1988; Stevens—Hanson, 1994). A theoretical study taking into account not only the kinetic

and frictional terms of the glottal impedance but also the glottal inductance revealed significant first formant bandwidth increases at glottal areas as small as  $0.02 \text{ cm}^2$ .

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### References

- Ananthapadmanabha, T. V.  
 1984 "Acoustic analysis of voice source dynamics", *STL-QPSR* 2-3/1984, Royal Institute of Technology, 1-24.
- Båvegård, M.—G. Fant  
 1995 "Interactive voice source modelling". *Proc. ICPhS-95*, Stockholm, 2: 634-637.
- Bickley, C. C.—K. N. Stevens  
 1986 "Effects of a vocal tract constriction on the glottal source: Experimental and modelling studies", *J. Phonetics* 14: 373-382.
- Blomberg, M.  
 1994 "Training production parameters of context-dependent phones for speech recognition", *STL-QPSR* 1/1994, Royal Institute of Technology, 59-90.
- Fant, G.  
 1959 *Acoustic Analysis and Synthesis of Speech with Applications to Swedish*, Ericsson Technics No. 1 /1959.  
 1960 *Acoustic Theory of Speech Production*, Mouton 1960. (Now available from Walter de Gruyter, Berlin).  
 1979 "Glottal source and excitation analysis". *STL-QPSR*, 1/1979, Royal Institute of Technology, 85-107.  
 1980 "Voice source dynamics", *STL-QPSR* 2-3/1980, Royal Institute of Technology, 17-37.  
 1982 "Preliminaries to the analysis of the human voice source", *STL-QPSR* 4/1982, Royal Institute of Technology, 1-27.

- 1986 "Glottal flow: models and interaction", *J. Phonetics*, 14: 393-399.
  - 1993 "Some problems in voice source analysis", *Speech Communication* 13: 7-22.
  - 1995 "The LF-model revisited. Transformations and frequency domain analysis", *STL-QPSR* 2-3/1995, Royal Institute of Technology, 119-156.
- Fant, G.—A. Kruckenberg
- 1994 "Notes on stress and word accent in Swedish", *Proc. International Symposium on Prosody*, 1994, Yokohama. (Also published in *STL-QPSR* 2-3/1994, Royal Institute of Technology, 125-144.)
  - 1995 "The voice source in prosody", *Proc. ICPhS-95*, Stockholm, 2: 622-625.
- Fant, G.—Q. Lin
- 1988 "Frequency domain interpretation and derivation of glottal flow parameters", *STL-QPSR* 2-3/1988, Royal Institute of Technology, 1-21.
- Fant, G.—J. Liljencrants—Q. Lin
- 1985 "A four-parameter model of glottal flow", *STL-QPSR* 4/1985, Royal Institute of Technology, 1-13.
- Fant, G.—A. Kruckenberg—J. Liljencrants—M. Båvegård
- 1994 "Voice source parameters in continuous speech. Transformation of LF-parameters", *Proc. ICSLP-94*, Yokohama, 1451-1454.
- Gauffin, J.—J. Sundberg
- 1980 "Data on the glottal voice source behaviour in vowel production", *STL-QPSR* 2-3/1980, Royal Institute of Technology, 61-70.
- Gobl, C.
- 1988 "Voice source dynamics in connected speech," *STL-QPSR* 1/1988, Royal Institute of Technology, 123-159.
- Gobl, C.—A. Ní Chasaide
- 1988 "The effects of adjacent voiced/voiceless consonants on the vowel voice source: a cross language study", *STL-QPSR* 2-3/1988, Royal Institute of Technology, 23-59.
- Hanson, H.
- 1995 "Individual variations in glottal characteristics of female speakers," *Proc. ICASSP 95*.
- Karlsson, I.
- 1988 "Glottal waveform parameters for different speaker types", *Proc. Speech 88, 7th FASE symposium*, Edinburgh, 69-72.

- 1990 "Voice source dynamics for female speakers", *Proc. ICSLP* Kobe, 69-72.
  - 1992 "Consonants for female speech synthesis", *Proc. ICSLP*, Banff 491-494.
- Karlsson, I.—L. Neovius
- 1993 "Speech synthesis experiments with the GLOVE synthesiser". *Proc. Eurospeech 93*, 925-928.
- Klatt, D.—L. Klatt,
- 1990 "Analysis, synthesis and perception of voice quality variations among female and male talkers", *J. Acoust. Soc. Am.* 87, 820-857.
- Lin, Q.
- 1990 *Speech Production Theory and Articulatory Speech Synthesis*, Ph.D. thesis, Dept. of Speech Communication and Music Acoustics, KTH, Stockholm.
- Sluijter, A.
- 1995 *Phonetic correlates of stress and accent*, Ph.D. thesis, Holland Institute of Generative Linguistics, Dordrecht: ICG Printing.
- Stevens, K. N.
- 1994 "Prosodic influences on glottal waveform: Preliminary data", *International Symposium on Prosody*, 1994, Yokohama, 53-63.
- Stevens, K. N.—M. Hanson
- 1994 "Classification of glottal vibration from acoustic measurements", in: O. Fujimura—M. Hirano (ed.), *Vocal Fold Physiology 1994*, San Diego: Singular Publ. Group, 147-170.